

LIMITING DISTRIBUTIONS OF THE ELEMENTARY SYMMETRIC FUNCTIONS OF SOME RANDOM MATRICES

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SUMMARY

The limiting distributions of the elementary symmetric functions of the characteristic roots of the Wishart matrix are derived. It is shown that the same theorem can be used for the noncentral Wishart matrix if the noncentrality parameter is of $O(1)$ or $O(n)$ with respect to the degrees of freedom n . The same technique is used in deriving the nonnull limiting distributions of the elementary symmetric functions for the beta matrix in the MANOVA case and the matrix appropriate for testing independence between two sets of variables.

1. Introduction

The asymptotic distributions that we are deriving will be limiting distributions, i.e. when the sample size $N \rightarrow \infty$. The method is based on a lemma given by Fujikoshi (1975).

Since we are only considering the limiting case, we state the lemma only to the necessary order terms.

Lemma 1.1

Let

$$(1.1) \quad M = \theta + \epsilon V + O(\epsilon^2).$$

where $\epsilon = N^{-1/2}$ *and* V *are* $O(1)$ *with respect to* N .

Let $\theta = \text{diag}(\theta_1, \dots, \theta_p)$, $\theta_1 \geq \dots \geq \theta_p$ *such that*

$$\theta_1 = \dots = \theta_{b_1} = \lambda_1, \theta_{b_1+1} = \dots = \theta_{b_1+b_2} = \lambda_2, \dots$$

$$\theta_{p-b_h+1} = \dots = \theta_p = \lambda_h$$

where $\lambda_1 > \dots > \lambda_h$ *and let* θ *and* v *be partitioned as*

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$$\theta = \begin{pmatrix} \lambda_1 I_{b_1} & o & \dots & o \\ o & \lambda_2 I_{b_2} & \dots & o \\ & \dots & \dots & \dots \\ o & o & \dots & \lambda_h I_{b_h} \end{pmatrix}, \quad v = \begin{pmatrix} V_{11} & \dots & V_{1h} \\ & \dots & \dots \\ V_{h1} & \dots & V_{hh} \end{pmatrix} \begin{pmatrix} b_1 \\ \dots \\ b_h \end{pmatrix}$$

Let J_α be the set of integers $b_1 + \dots + b_{\alpha-1} + 1, \dots, b_1 + \dots + b_\alpha$ then for $d_1 > \dots > d_{p'}$ the characteristic roots of M , d_j , $j \in J_\alpha$ are equal to the characteristic roots of

$$(1.2) \quad z_\alpha = \lambda_\alpha I_{b_\alpha} + \epsilon V_{\alpha\alpha} + O(\epsilon^2).$$

Any special cases of this lemma are easy to construct and will not be given in detail.

The following lemmas will be useful in deriving the distributions. The proof of the first one is omitted since it is easy to prove.

Lemma 1.2 If M is in the form of (1.1) and $\theta_1 > \dots > \theta_{p'}$ then

$$(1.3) \quad \text{tr}_j M = \text{tr}_j \theta + \epsilon v_j + O(\epsilon^2)$$

where

$$v_j = \sum_{i_1 < \dots < i_j} \{ z_{i_1}^{\theta_{i_1}} \dots z_{i_j}^{\theta_{i_j}} + \dots + \theta_{i_1} \dots \theta_{i_{j-1}} z_{i_j} \}.$$

and the characteristic function of

$$T = \frac{1}{\epsilon} (\text{tr}_j M - \text{tr}_j \theta)$$

is given by

$$(1.4) \quad \phi_T(t) = E \exp(itv_j) \{1 + O(\epsilon)\}$$

It is interesting to note that if ϵ is replaced by $(m-2\gamma)^{-1/2}$ where γ is some fixed constant, then the characteristic function of

$$(1.5) \quad T_{\gamma} = \sqrt{m} (\text{tr}_j M - \text{tr}_j \theta)$$

can be written as

$$(1.6) \quad \theta_{T_{\gamma}}(t) = \theta_T \left(\sqrt{\frac{m}{m-2\gamma}} t \right)$$

$$(1.6) \quad = E \exp(it \nu_j) \{1 + O(m^{-1/2})\}$$

If all the elements of θ are distinct, then $b_1 = b_2 = \dots = b_h = 1$ and $h = p$. If all the elements are the same, say λ , then $h = 1$ and $b_1 = p$.

Two lemmas that will be used are the following:

Lemma 1.3

Let

$$(1.7) \quad C_1 = \sum_{i_1=1}^p \theta_{i_1}^2 \left(\sum_{\substack{i_2 < \dots < i_j \\ \neq i_1}} \theta_{i_2} \dots \theta_{i_j} \right)^2$$

then

$$(1.8) \quad C_1 = j \text{tr}_j \theta^2 + \sum_{k=1}^j (-1)^{k+1} 2^{j+(-1)^k k} \text{tr}_{j+k} \theta \text{tr}_{j-k} \theta.$$

Proof:

(1.7) can be partitioned into the sum of terms containing (i) the product of j characteristic roots squared, (ii) the product of $(j-1)$ ch. roots squared times two roots, (iii) the product of $j-2$ ch. roots squared times three roots, etc. It is obvious that proceeding in this way the last terms contain at least one ch. root squared and the other $(j-1)$ of linear order. It is further clear that the total number of terms in (1.7) is $p \binom{p-1}{j-1}^2$. It is also true that there are

$$(1.9) \quad p \binom{p-1}{i-1} \binom{p-j}{j-s} \binom{j-1}{s-1}$$

terms with s roots squared. It is easy to check

that

$$\sum_{s=0}^j p \binom{p-1}{j-1} \binom{p-j}{j-s} \binom{j-1}{s-1} = p \binom{p-1}{j-1}^2.$$

From (1.8) however there are

$$(1.10) \quad \sum_{k=1}^j (-1)^{k+1} 2^{j+(-1)^k k} \binom{p}{j-k} \binom{j-k}{s} \binom{p-j+k}{j+k-s}$$

terms with s roots squared. This follows since $\text{tr}_k \theta \text{tr}_i \theta$, $i \leq k$, has $\left(\binom{p}{i} \binom{i}{s} \binom{p-i}{k-s} \right)$ terms containing s quadratic roots. The equality of (1.10) and (1.9) follows easily (see Lombard and De Waal (1976) for the proof of a more general lemma).

Lemma 1.4

Let

$$(1.11) \quad C_2 = \sum_{i_1=1}^p \theta_{i_1} \left(\sum_{\substack{i_2 < \dots < i_j \\ \neq i_1}} \theta_{i_2} \dots \theta_{i_j} \right)^2$$

then

$$(1.12) \quad C_2 = \sum_{k=1}^j (2j-2k+1) \text{tr}_{2j-k} \theta \text{tr}_{k-1} \theta$$

Proof:

The total number of terms in (1.11) is

$$p \binom{p-1}{j-1}^2$$

which can partitioned into

$$(1.13) \quad p \binom{p-1}{j-1} \binom{p-j}{j-s-1} \binom{j-1}{s}, \quad s = 0, \dots, j-1$$

terms containing s ch. roots squared.

On the other hand (1.12) has

$$(1.14) \quad \sum_{k=1}^j (2j-2k+1) \binom{p}{k-1} \binom{k-1}{s} \binom{p-k+1}{2j-k-s}$$

terms containing s ch. roots squared. The equality of (1.14) and (1.13) is not hard to prove. A more general proof can be found in Lombard and De Waal (1976).

The purpose of this paper is to derive the limiting distributions of the elementary symmetric functions of the characteristic roots of M , i.e.

$$\text{tr}_j M = \sum_{i_1 < \dots < i_j} d_{i_1} \dots d_{i_j}$$

by assuming M as a random matrix in various cases. Specifically the following cases will be considered.

- (i) M is a Wishart matrix (central or noncentral),
- (ii) M is a beta matrix of the second type — MANOVA case.
- (iii) M is conditionally a beta matrix of the second type — the case of independence test.

We shall treat the Wishart case in section 2, the MANOVA case in section 3 and the test for independence case in section 4.

2. Limiting distribution of the esfs of the characteristic roots of the Wishart Matrix

If we assume $A(p \times p)$ is distributed $W(\Sigma, n)$, then in the limit when $n \rightarrow \infty$, we may write (cf. Fujikoshi (1975))

$$A = n\Sigma + \sqrt{n}W,$$

where $W = W'$ is distributed normally independent of n with $E(W) = 0$ and some covariance structure. Since Σ is not diagonal, we make an orthogonal transformation on it to get the form (1.1). Let $H(p \times p)$ be orthogonal such that

$$H\Sigma H' = \text{diag.}(\theta_1, \dots, \theta_p), \theta_1 > \dots > \theta_p$$

Let

$$(2.1) \quad M = \frac{1}{n} HAH' = \theta + \frac{1}{\sqrt{n}} V$$

where $V = HWH'$. Clearly $V = V'$ will have a limiting normal distribution with $E(V) = 0$. It is easy to see that the characteristic function of $V_{11}, 2V_{12}, 2V_{13}, \dots, V_{pp}$ is given by

$$(2.2) \quad \phi_V(T) = E \text{etr}(iVT) \\ = \text{etr}-(T)^2$$

Therefore

$$(2.3) \quad \begin{aligned} \text{Var}(V_{ii}) &= 2\theta_i^2 \\ \text{Var}(V_{ij}) &= \theta_i \theta_j \\ \text{Cov}(V_{ij}, V_{ek}) &= 0 \quad \begin{aligned} &i \neq e \text{ or } k \\ &\text{or } j \neq e \text{ or } k \end{aligned} \end{aligned}$$

M given in (2.1) is now of the form (1.1) and it may be noted that the characteristic roots of M are also the characteristic roots of $\frac{1}{n}A$.

If all the roots of Σ are the same, say $\Sigma = I_p$, then M is equal to

$$(2.4) \quad M = I + \frac{1}{\sqrt{n}} V$$

for $n \rightarrow \infty$,

and then

$$(2.5) \quad \text{tr}_j M = \binom{p}{j} + \frac{1}{\sqrt{n}} \binom{p-1}{j-1} \text{tr} V + \frac{1}{n} \binom{p-2}{j-2} \text{tr}_2 V \\ + \dots + \frac{1}{n^{j/2}} \text{tr}_j V$$

Hence from (2.4), the characteristic function of

$$(2.6) \quad T_1 = \sqrt{n} (\text{tr}_j M - \binom{p}{j})$$

is given by

$$(2.7) \quad \phi_{T_1}(t) = E \exp \left(i t \left(\binom{p-1}{j-1} \text{tr} V \right) \{1 + O(n^{-1/2})\} \right)$$

We are now in a position to derive the asymptotic distributions of the elementary symmetric functions of the characteristic roots of the central Wishart matrix.

Theorem 2.1

If A is distributed $W(I, n)$ then the limiting distribution of

$$(2.8) \quad T_1 = \sqrt{\binom{n}{2p}} \left\{ \text{tr}_j \frac{1}{n} A - \binom{p}{j} \right\} / \binom{p-1}{j-1}$$

is $N(0; 1)$

Proof:

From (2.7)

$$\lim_{n \rightarrow \infty} \phi_{T_1}(t) = \exp \{-t^2 p \binom{p-1}{j-1}^2\}$$

which is the characteristic function of normal variable with mean 0 and variance $2 \binom{p-1}{j-1}^2 p$.

Theorem 2.2

If A is distributed $W(\Sigma, n)$ and if $\theta_1 > \dots > \theta_p$ are the characteristic roots of Σ , then the limiting distribution of

$$T = \frac{\sqrt{n}}{\sqrt{2C_1}} \{tr_j \frac{1}{n} A - tr_j \theta\}$$

is $N(0; 1)$. C_1 is given in (1.8)

Proof:

From (1.4) with $\epsilon = n^{-1/2}$

$$\lim_{n \rightarrow \infty} \phi_T(t) = \exp[-t^2 C_1]$$

where C_1 is given in (1.7) and (1.8),

which is the characteristic function of a normal variable with mean 0 and variance $2C_1$.

If we consider A as a noncentral Wishart matrix, namely distributed $W(\Sigma, n, \Omega)$, then it is known that for

$$(i) \quad \Omega = O(n) = n\xi, \text{ say,}$$

then for $n \rightarrow \infty$

$$(2.10) \quad \frac{1}{n} A = \Sigma(I + \xi) + \frac{1}{\sqrt{n}} W$$

and W has the same distribution as that given in the beginning of this section.

$$(ii) \quad \Omega = O(1)$$

then for $n \rightarrow \infty$

$$(2.11) \quad \frac{1}{n} A = \Sigma + \frac{1}{\sqrt{n}} W.$$

Theorem 2.2 is therefore applicable in both cases by assuming θ as the characteristic roots of $\Sigma(I + \xi)$ and Σ respectively.

3. Limiting distributions for the esf's of the characteristic roots of matrices in the MANOVA case

In the MANOVA case we end up with two matrices S_E and S_H , called the variance

due to error and the variance due to hypothesis, respectively. $S_E(\text{pxp})$ is distributed $W(1, n)$ independent of $S_H(\text{pxp})$ distributed as $W(1, q, \Omega)$ in the nonnull case ($n = N - b$, b some fixed constant, $q \gg p$). In the null case $\Omega = 0$. The matrix we are interested in is $S_H S_E^{-1}$ and we shall derive the limiting distribution of $\text{tr}_j S_H S_E^{-1}$ in the nonnull case, $\Omega = O(n) = n\theta$, say. We shall however briefly give an indication why the null case $\Omega = 0$ and the nonnull case $\Omega = O(1)$ are not treated here.

Using (2.4) $n S_H S_E^{-1}$ can be written in the limit $n \rightarrow \infty$ as

$$(3.1) \quad n S_H S_E^{-1} = S_H \left(I + \frac{1}{\sqrt{n}} V \right)^{-1} \\ = S_H \left(I - \frac{1}{\sqrt{n}} V + n^{-1} V^2 + \dots \right)$$

There exists an $C(\text{pxp})$ orthogonal such that

$$C S_H C' = \text{diag}(S_1, \dots, S_p) = C_S \quad \text{say}$$

where $S_1 > \dots > S_p$ are the characteristic roots of S_H .

Since the characteristic roots of $C S_H S_E^{-1} C'$ are the same as $S_H S_E^{-1}$, we may write in the limit as $n \rightarrow \infty$

$$(3.2) \quad M = C_S - \frac{1}{\sqrt{n}} W + \frac{1}{n} W^2 + \dots,$$

where $M = n C S_H S_E^{-1} C'$ and $W = W' = C' V C$ is distributed normally with mean zero and $\text{Var}(W_{ij}) = 2$, $\text{Var}(W_{jj}) = 1$. The covariances are all zero. This follows because S_H and S_E are independently distributed and the characteristic function of W conditional on C does not contain C .

Hence the characteristic function of M is given by

$$(3.3) \quad \phi_M(t) = E \exp(i t \text{tr}_j S_H) \{1 + O(n^{-1/2})\}.$$

This expectation is however unknown.

Nonnull case $\Omega = O(n) = n\theta$

Assuming S_H is distributed $W(1, q, \Omega)$ and $\Omega = n\theta$, say, we may write

$$(3.4) \quad S_H = (X + \sqrt{n} \theta^{1/2}) (X + \sqrt{n} \theta^{1/2})' + Y Y',$$

i.e.

$$(3.5) \quad \frac{1}{n} S_H = \theta + \frac{1}{\sqrt{n}} (X \theta^{1/2} + \theta^{1/2} X') + \frac{1}{n} W,$$

where $W = XX' + YY'$ and $X(px \times p)$ and $Y(px \times (q-p))$ have independent standard normal elements. Hence using (2.4) and (3.5)

$$(3.6) \quad S_H S_E^{-1} = \theta + \frac{1}{\sqrt{n}} D + \dots,$$

$$\text{where } D = X \theta^{\frac{1}{2}} + \theta^{\frac{1}{2}} X' - \theta V.$$

The following Theorem can now be proved:

Theorem 3.1

If $S_H S_E^{-1}$ can be written in the form (3.6), then the limiting distribution of

$$(3.7) \quad T = \sqrt{n} (tr_j S_H S_E^{-1} - tr_j \theta)$$

is $N(0; 2C_1 + 4C_2)$ where C_1 and C_2 are given in (1.8) and (1.12) respectively.

Proof:

From (1.3) and (1.4) the characteristic function of T can be written as

$$(3.8) \quad \theta_T(t) = E \exp(it v_j) + O(n^{-\frac{1}{2}})$$

where

$$v_j = \sum_{i_1 < \dots < i_j} \{ Z_{i_1} \theta_{i_1} \dots \theta_{i_j} + \dots + \theta_{i_1} \dots \theta_{i_{j-1}} z_{i_j} \}$$

$$\text{and} \quad z_\alpha = X_{\alpha\alpha} \theta_\alpha^{\frac{1}{2}} + \theta_\alpha^{\frac{1}{2}} X_{\alpha\alpha} - \theta_\alpha V_{\alpha\alpha}$$

The variance of Z_α is

$$(3.9) \quad \text{Var}(Z_\alpha) = 4\theta_\alpha + 2\theta_\alpha^2.$$

Therefore (3.8) becomes, by using (1.8) and (1.12),

$$(3.10) \quad \theta_T(t) = \exp[-t^2(2C_2 + C_1)] + O(n^{-\frac{1}{2}})$$

where C_1 and C_2 are given in (1.8) and (1.12) respectively.

4. Test for independence

From Fujikoshi (1975)

$$(4.1) \quad G = \theta + \frac{1}{\sqrt{n}} G + \dots$$

where G has ch. roots $g_1 > \dots > g_p$ and

$$\theta = \text{diag}(\theta_1, \dots, \theta_p), \theta_1 \geq \dots \geq \theta_p, g_j = \frac{r_j^2}{1-r_j^2}$$

and $\theta_j = \frac{\rho_j^2}{1-\rho_j^2}$ where $r_1^2 > \dots > r_p^2$ are the sample canonical correlations between p com-

ponents and q components ($p \leq q$) of a $(p+q)$ variate normal based on a sample of $(n+1)$ observations.

$\rho_1^2 \geq \dots \geq \rho_p^2$ are the population canonical correlations. Furthermore

$G = D + H$ where D is defined in (3.6) and

$$(4.2) \quad H = \theta^{1/2} L \theta^{1/2}.$$

$V = V'$ and $L = L'$ have the same limiting normal distributions and are independent and $X: p \times p$ is independent of V and L with independent standard normal elements.

The following theorem can now be proved:

Theorem 4.1

If G is defined as in (4.1) then the limiting distribution of

$$(4.3) \quad T = \sqrt{n} (\text{tr}_f G - \text{tr}_f \theta)$$

is $N(0; 4(C_2 + C_1))$ where C_1 and C_2 are given in (1.8) and (1.12) respectively.

Proof:

From (1.3) and (1.4) the characteristic function of T can be written as

$$\phi_T(t) = E \exp(itv_j) + O(n^{-1/2})$$

where $v_j = \sum_{i_1 \leq \dots \leq i_j} \{Z_{i_1} \theta_{i_2} \dots \theta_{i_j} + \dots + \theta_{i_1} \dots \theta_{i_{j-1}} Z_{i_j}\}$

and

$$Z_{\alpha} = X_{\alpha\alpha}\theta^{1/2} + \theta^{1/2}X_{\alpha\alpha} - \theta_{\alpha}V_{\alpha\alpha} + \theta^{1/2}L_{\alpha\alpha}\theta^{1/2}.$$

Thus

$$\text{Var}(Z_{\alpha}) = 4\theta_{\alpha} + 4\theta_{\alpha}^2.$$

Hence by using (1.8) and (1.12) the characteristic function of T can be written as

$$\phi_T(t) = \exp(-2t^2(C_2 + C_1)) + O(n^{-1/2})$$

and this completes the proof.

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